

LOCARNO MAGADINO AP, Switzerland

WMO: 067620

Lat: 46.166N

Lon: 8.882E

Elev: 650

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	23.8	3.9	6.9	39.7	8.6	8.8	40.9	21.1	45.8	18.3	46.0	2.5	90	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	87.2	71.3	84.9	69.9	82.7	68.7	73.4	84.3	71.8	82.1	70.4	80.1	6.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	112.5	78.6	68.3	106.5	76.8	67.0	101.7	75.4	37.3	84.5	35.9	82.2	34.7	80.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	13.4	10.7	DB	17.4	90.2	4.7	2.5	14.0	92.0	11.2	93.5	8.6	94.9	5.1	96.7	(4)
(5)			WB	16.5	75.9	4.5	1.7	13.3	77.1	10.7	78.0	8.2	79.0	4.9	80.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	35.8	39.2	47.2	53.9	61.5	68.7	72.0	70.7	63.3	54.5	44.5	36.4	(6)	
	DBStd	13.71	4.75	4.89	5.43	5.29	5.04	5.03	4.28	4.12	4.75	5.16	5.64	5.23	(7)	
	HDD50	1498	440	302	117	21	1	0	0	0	0	17	180	420	(8)	
	HDD65	4582	904	721	551	333	131	20	4	6	87	324	616	885	(9)	
	CDD50	2984	1	1	31	140	356	562	681	642	398	158	14	0	(10)	
	CDD65	591	0	0	0	1	21	133	220	182	34	0	0	0	(11)	
	CDH74	4638	0	0	5	28	211	1083	1802	1297	205	7	0	0	(12)	
	CDH80	1154	0	0	0	3	25	275	509	322	19	1	0	0	(13)	
(14) Wind	WSAvg	3.5	3.0	3.3	4.1	4.2	4.1	4.0	4.0	3.6	3.3	2.9	2.9	2.7	(14)	
(15) Precipitation	PrecAvg	68.0	2.9	2.6	3.2	5.8	7.6	7.1	6.6	7.0	7.5	7.1	7.6	2.9	(15)	
	PrecMax	100.7	9.6	10.9	10.9	15.7	21.8	17.6	12.8	12.6	17.4	22.7	29.2	8.9	(16)	
	PrecMin	39.1	0.0	0.2	0.2	0.5	2.8	1.1	1.2	0.8	0.8	0.4	0.1	0.0	(17)	
	PrecStd	14.9	2.4	2.4	2.5	3.8	4.1	3.7	3.1	2.9	5.1	4.8	6.9	2.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.3	61.5	72.5	77.3	83.3	89.4	90.0	90.1	82.5	73.7	62.6	56.7	(19)	
		MCWB	44.3	46.0	52.5	59.6	65.5	71.4	72.9	73.5	68.1	60.8	50.5	42.5	(20)	
	2%	DB	50.8	55.7	66.5	73.2	79.1	86.0	87.3	86.2	79.1	69.4	58.5	50.7	(21)	
		MCWB	40.0	43.8	50.7	57.3	63.5	70.5	71.3	71.3	65.8	59.4	49.9	43.0	(22)	
	5%	DB	47.2	52.4	62.9	70.0	76.0	83.3	85.2	83.5	76.2	66.6	56.0	47.6	(23)	
		MCWB	39.1	42.2	49.3	55.2	61.7	69.1	70.3	69.6	64.0	58.0	49.6	42.1	(24)	
	10%	DB	44.0	49.3	59.6	66.6	73.1	80.4	82.9	81.0	73.4	63.9	53.6	45.1	(25)	
		MCWB	38.0	40.6	47.8	53.5	59.9	67.1	68.8	68.4	62.3	57.1	49.1	40.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	48.3	54.6	61.5	67.7	74.6	75.6	74.7	70.2	63.8	55.3	49.0	(27)	
		MCDB	54.1	57.3	67.2	74.2	79.8	86.3	86.1	86.1	79.2	69.4	57.7	50.8	(28)	
	2%	WB	43.0	45.5	52.6	58.6	65.6	72.1	73.6	72.9	68.1	62.0	53.4	45.8	(29)	
		MCDB	47.5	52.6	62.8	70.4	76.4	83.2	84.4	83.6	75.9	66.5	55.9	47.7	(30)	
	5%	WB	40.9	43.7	51.1	56.7	63.6	70.4	71.9	71.4	66.5	60.3	51.8	43.5	(31)	
		MCDB	44.9	49.9	59.8	66.7	73.2	80.9	82.3	81.2	73.2	64.2	54.1	46.3	(32)	
	10%	WB	39.0	42.1	49.3	55.0	61.8	68.7	70.5	70.0	64.7	58.7	50.1	41.4	(33)	
		MCDB	42.7	47.4	57.4	63.5	70.0	78.0	80.3	78.6	70.7	62.3	52.4	44.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	17.7	20.3	19.3	18.3	18.9	19.8	18.8	17.8	16.1	14.5	14.6	(35)	
		MCDBR	20.4	23.8	27.7	27.6	24.6	23.3	22.7	22.6	21.9	20.6	18.7	17.9	(36)	
	5% WB	MCWBR	12.3	14.0	14.7	13.7	11.7	10.9	10.2	10.4	10.9	11.7	11.3	11.3	(37)	
		MCWBR	17.3	21.0	24.1	23.7	21.2	21.3	20.7	20.5	18.5	15.8	12.1	14.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.319	0.356	0.419	0.489	0.502	0.518	0.502	0.491	0.468	0.430	0.362	0.314	(40)	
		taud	2.274	2.178	2.056	1.937	1.983	1.999	2.058	2.097	2.115	2.169	2.289	2.322	(41)	
	Ebn at Noon		241	252	251	244	246	242	245	241	234	223	222	229	(42)	
		Edn at Noon	26	35	45	56	55	55	51	47	43	35	26	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		495	784	1182	1421	1597	1866	1878	1614	1241	791	483	412	(44)	
	RadStd		64	98	164	207	149	145	146	127	109	103	77	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+1.31	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (114 neighbors)	N/A	N/A	N/A	+1.30	+0.72	+0.56	N/A	N/A	+106	+29

Nomenclature: See separate page